



FleetWise
BY RAPID RISE AI

Build checklist and functional requirements

The complete, itemised specification of everything FleetWise must do for production: every feature, how features interact, the data it holds, the metrics it must report, and the acceptance criteria that define "done".

Engineering reference & gap-analysis baseline · Rapid Rise AI · 2026 · v1.0

How to use this document. Every requirement has a stable ID (e.g. FR-3.2.4). Use these IDs to track implementation status, write tickets, and drive a gap analysis against the current codebase. A requirement is only "done" when its acceptance criteria are met.

Priority key: **P0** launch-critical **P1** important **P2** later

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1 · Product scope & principles

- **FR-1.1** **P0** Manage every asset on a farm in one system: vehicles, implements and equipment.
- **FR-1.2** **P0** Every data-capture action (service, fuel, fault) completable in under 30 seconds on mobile.
- **FR-1.3** **P0** Full functionality offline; nothing critical requires a live connection.
- **FR-1.4** **P1** Every record is auditable: who created/changed it, when, and from where.
- **FR-1.5** **P0** Multi-farm / multi-site support under one account, with data isolation per site.

2 · User roles & permissions

- **FR-2.1** **P0** Role-based access control with at least: Owner, Manager, Operator/Driver, Workshop/Mechanic, External Contractor.
- **FR-2.2** **P0** Unlimited users per account, regardless of plan.
- **FR-2.3** **P1** Per-role visibility: operators see only their assigned assets; contractors see only assigned jobs.
- **FR-2.4** **P1** Invite/deactivate users; enforce least-privilege by default.
- **FR-2.5** **P2** Custom roles with granular permission toggles.

3 · Asset register

- **FR-3.1** **P0** Create/edit/archive assets of any type (tractor, bakkie, truck, implement, pump, generator, other).
- **FR-3.2** **P0** Per-asset fields: make, model, year, registration, VIN/serial, purchase date & price, finance details, photos.
- **FR-3.3** **P0** Live status per asset: In use, Available, In workshop, Out of service.
- **FR-3.4** **P1** Group/filter by type, location, cost centre, department.
- **FR-3.5** **P0** Current engine-hour and odometer reading per asset, updated on each relevant log.
- **FR-3.6** **P1** Assign a default operator/driver to each asset.
- **FR-3.7** **P1** Bulk import assets (CSV/Excel) during onboarding.

4 · Maintenance & servicing

- **FR-4.1** **P0** Complete, dated service/maintenance history per machine (fuel, parts, labour, breakages, repairs).
- **FR-4.2** **P0** Schedule services by engine hours, kilometres AND calendar time, whichever is due first.
- **FR-4.3** **P0** Recurring service intervals per machine (e.g. every 250 engine hours).
- **FR-4.4** **P0** Due / due-soon / overdue states computed automatically from latest readings.
- **FR-4.5** **P1** Attach photos and notes to every service record.
- **FR-4.6** **P1** Job cards: plan a service, mark in-progress, complete, and record who did it.
- **FR-4.7** **P2** Warranty period tracking per machine with expiry reminders (links to §13).

5 · Service kits & parts

- **FR-5.1** **P0** Pre-loaded service kit per machine: exact engine oil, gearbox oil and filter part numbers.
- **FR-5.2** **P1** Parts catalogue with part number, description, supplier and typical cost.
- **FR-5.3** **P1** Parts consumed on a service auto-append to that machine's cost and history.
- **FR-5.4** **P2** Optional parts inventory/stock levels with low-stock alerts.

6 · Fuel & consumables

- **FR-6.1** **P0** Log fuel entries per asset (litres, cost, date, engine hours/odo, operator).
- **FR-6.2** **P0** Compute consumption (L/hr or L/100km) and trend per asset.
- **FR-6.3** **P0** Anomaly detection: flag fuel entries that deviate from the asset's baseline (possible leak/theft).
- **FR-6.4** **P2** Import fuel-card transactions and reconcile against logged entries (links to §17).

7 · Fault & issue reporting

- **FR-7.1** **P0** Operators log a fault against an asset in seconds, with description, severity and photos.
- **FR-7.2** **P0** Each fault captures asset, reporter, timestamp and (if available) location.
- **FR-7.3** **P1** Fault lifecycle: Open → Acknowledged → In progress → Resolved, with assignee.
- **FR-7.4** **P1** A fault can be converted into a job card / service (links to §4).
- **FR-7.5** **P1** Reporting a fault can flip asset status to Out of service (links to §3).

8 · Contractor & mechanic portal

- **FR-8.1** **P0** External mechanics get scoped access to log work only on assigned assets/jobs.
- **FR-8.2** **P0** Upload quotes, invoices and job photos against an asset.
- **FR-8.3** **P1** Update job status; owner/manager notified on change (links to §14).
- **FR-8.4** **P1** Uploaded invoice amounts flow into that asset's cost record (links to §10).

9 · QR codes & field capture

- **FR-9.1** **P0** Generate a unique QR code per asset; printable as a weatherproof sticker.
- **FR-9.2** **P0** Scanning a QR opens that asset and offers quick actions: log service, log fuel, report fault.
- **FR-9.3** **P1** Scan-to-log works offline and queues for sync (links to §15).
- **FR-9.4** **P2** Re-issue/replace a QR code if a sticker is damaged, preserving asset link.

10 · Costs & true cost of ownership

- **FR-10.1** **P0** Every cost (fuel, parts, labour, external invoice, finance/interest) attributed to its asset.
- **FR-10.2** **P0** True cost of ownership per asset (purchase + finance + fuel + maintenance + repairs).
- **FR-10.3** **P0** Cost per engine hour and per kilometre, computed and displayed.
- **FR-10.4** **P1** Budgets per asset/category with budget-vs-actual tracking.
- **FR-10.5** **P2** Repair-vs-replace indicator when lifetime cost crosses a threshold of asset value.

11 · Dashboard & reporting

- **FR-11.1** **P0** Dashboard: machines due/overdue, total maintenance spend, assets tracked, cost by machine.
- **FR-11.2** **P0** "Breaks most often" analysis: parts/machines with the highest repeat repairs.
- **FR-11.3** **P1** Selectable period ranges and per-site/per-group filtering.
- **FR-11.4** **P1** Export reports to Excel/CSV and PDF.
- **FR-11.5** **P2** Custom/scheduled reports emailed on a cadence.

12 · Voice control AI assistant

- **FR-12.1** **P0** Voice control in English and Afrikaans to log services/faults and query status hands-free.
- **FR-12.2** **P1** Assistant confirms the action back and asks for missing detail before committing.
- **FR-12.3** **P1** Voice actions respect the same permissions and validation as manual entry.
- **FR-12.4** **P2** Graceful fallback to manual entry when offline or on recognition failure.

13 · Compliance (AARTO, audits, docs)

- **FR-13.1** **P0** Record which driver operated each vehicle and when, to support AARTO driver nomination.
- **FR-13.2** **P1** AARTO fine workflow: capture a fine, identify driver from usage log, track nomination deadline.
- **FR-13.3** **P1** Licence/renewal tracking with reminders.
- **FR-13.4** **P1** Export-audit pack for GLOBALG.A.P. / SIZA; vehicle-sale and warranty-claim document packs.

14 · Notifications & reminders

- **FR-14.1** **P0** Service-due, overdue and licence-expiry reminders via in-app, push and WhatsApp.
- **FR-14.2** **P1** Notify relevant roles on fault reported, job status change, and fuel anomaly.
- **FR-14.3** **P2** Per-user notification preferences and quiet hours.

15 · Offline & sync

- **FR-15.1** **P0** All capture actions work offline and queue locally.
- **FR-15.2** **P0** Automatic sync when connectivity returns, with visible sync status.
- **FR-15.3** **P0** Deterministic conflict resolution (no silent data loss; last-writer + audit trail).
- **FR-15.4** **P1** Offline media (photos) cached and uploaded on reconnect.

16 · WhatsApp & mobile capture

- **FR-16.1** **P0** Log a service or fault via WhatsApp, resolving to the correct asset and user.
- **FR-16.2** **P1** WhatsApp reminders and confirmations tie back to the same records as the app.

- **FR-16.3** **P0** Responsive mobile UI; touch targets $\geq 44\text{px}$; works on low-end Android.

17 · Integrations & API

- **FR-17.1** **P1** Fuel-card import; GPS/telematics tracker feed for engine hours & location.
- **FR-17.2** **P2** Accounting export/integration (e.g. Sage/Xero) for costs.
- **FR-17.3** **P2** Public REST API with token auth for custom integrations.

18 · Localisation (EN/AF)

- **FR-18.1** **P0** Full UI in English and Afrikaans, switchable per user, choice persisted.
- **FR-18.2** **P1** Rand currency, local date formats, and ZA units throughout.

19 · Billing, plans & entitlements

- **FR-19.1** **P0** Per-vehicle-per-month billing; unlimited users; four tiers (Essential/Professional/Complete/Done-For-You).
- **FR-19.2** **P0** Feature entitlements gated by plan (e.g. dashboard = Professional+, voice AI & AARTO = Complete+).
- **FR-19.3** **P1** Annual pre-pay (2 months free); asset-count-based pricing; no lock-in / easy export on cancel.
- **FR-19.4** **P2** Self-hosted licence deployment path.

20 · Feature interaction map

The system's value comes from how modules feed each other. Each of these data flows must exist end to end:

- A **fuel/service log** updates the asset's **engine hours/odometer** → recomputes **service-due** state → may fire a **reminder**.
- A **QR scan** or **voice command** or **WhatsApp message** all create the same underlying log records.
- A **fault report** can set asset status **Out of service**, spawn a **job card**, and notify the manager.
- A **contractor invoice** upload adds to the asset's **cost record** and **true cost of ownership**.
- Every cost event feeds the **dashboard metrics** and **cost-per-hour/km** calculations.
- Driver-to-vehicle **usage logs** power both **accountability** and **AARTO nomination**.
- Any capture offline queues and later **syncs**, after which all dependent metrics recompute.

21 · Data model summary

Core entities the system must persist (with relationships):

- **Organisation** → has many **Sites/Farms** → has many **Assets, Users**.

- **Asset** → has many ServiceRecords, FuelEntries, Faults, CostEntries, Documents, QRCode(1), ServiceSchedule(s), ServiceKit(1).
- **User** (role) → creates logs; **Operator** ↔ Asset assignment; **Contractor** ↔ Job assignment.
- **ServiceRecord** → references Parts (from Catalogue), Labour, Photos, performedBy.
- **Fault** → status lifecycle; optional → JobCard → ServiceRecord.
- **CostEntry** → type (fuel/parts/labour/invoice/finance) → rolls up to Asset TCO.
- **UsageLog** (driver, asset, time) → feeds accountability & AARTO. **Fine** → links UsageLog + deadline.
- **Subscription** → plan, asset count, entitlements, billing period.
- **SyncQueue** → pending offline mutations + conflict metadata. **AuditLog** → all changes.

22 · Non-functional requirements

- **NFR-1** **P0** Performance: key screens interactive < 2s on 3G / low-end Android.
- **NFR-2** **P0** Security: encrypted in transit & at rest; hashed credentials; per-tenant data isolation.
- **NFR-3** **P0** POPIA compliance for personal/driver data; documented retention & deletion.
- **NFR-4** **P1** Reliability: automated backups, documented restore, ≥ 99.5% uptime target.
- **NFR-5** **P1** Accessibility: legible in sunlight/high-contrast, large tap targets, screen-reader labels.
- **NFR-6** **P1** Observability: error logging, sync-failure alerts, usage analytics.
- **NFR-7** **P1** Scalability: thousands of assets per org without degradation.

23 · Metrics & stats to compute

The system must be able to derive and display all of the following:

- | | |
|--|--|
| • Machines due / overdue for service | • Fuel consumption (L/hr, L/100km) & trend |
| • Total maintenance spend (period) | • Fuel anomalies flagged |
| • Number of assets tracked | • Downtime per asset |
| • Cost by machine (ranked) | • Open vs resolved faults |
| • Parts/machines that break most often | • Budget vs actual |
| • Total accumulated cost per asset (TCO) | • Utilisation (hours used vs idle) |
| • Cost per engine hour | • Warranty/licence expiries upcoming |
| • Cost per kilometre | • AARTO nominations pending & deadlines |

24 · Production readiness gate

FleetWise is production-ready only when:

- All **P0** requirements are implemented, tested and pass their acceptance criteria.
- Every \$20 data flow works end to end across app, QR, voice and WhatsApp inputs.
- Offline capture + sync verified with conflict cases and no data loss.

- All §23 metrics compute correctly against seeded and real-shaped data.
- NFR security, POPIA, backup/restore and performance targets verified.
- Plan entitlements enforced; billing correct; EN/AF complete with no missing strings.